**3 Complex Analysis 2024**
Tutorial 9

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# Singularities

**Problem 37** Which of the following functions have a removable singularity at the given point $z_0$?

(a) $\dfrac{6-z-z^2}{2-z}$, $z_0 = 2$.

(b) $\dfrac{\sin z}{z^2-\pi^2}$, $z_0 = \pi$.

(c) $\dfrac{\sin z}{(z-\pi)^2}$, $z_0 = \pi$.

(d) $\dfrac{\sin z}{z^2+z}$, $z_0 = 0$.

**Solution.**

(a) We have

$$\frac{6-z-z^2}{2-z} = \frac{-(z-2)(z+3)}{-(z-2)} = z+3, \quad z \neq 2.$$

Thus, the limit

$$\lim_{z\to 2} \frac{6-z-z^2}{2-z} = \lim_{z\to 2}(z+3) = 5$$

exists, $z_0 = 2$ is a removable singularity.

(b) The limit

$$\lim_{z\to\pi} \frac{\sin z}{z^2-\pi^2} = \lim_{z\to\pi} \frac{\sin z}{(z-\pi)(z+\pi)} = \lim_{z\to\pi} \frac{\sin z - \sin \pi}{z-\pi} \lim_{z\to\pi} \frac{1}{z+\pi}$$
$$= (\sin z)'\big|_{z=\pi} \cdot \frac{1}{2\pi} = \cos \pi \cdot \frac{1}{2\pi} = -\frac{1}{2\pi}$$

exists, $z_0 = \pi$ is a removable singularity.

(c) Here we have

$$\frac{\sin z}{(z-\pi)^2} = \frac{\sin z - \sin \pi}{z-\pi} \cdot \frac{1}{z-\pi}.$$

As we have seen above, the first factor $\frac{\sin z - \sin \pi}{z - \pi}$ has a limit as $z \to \pi$, namely, $\lim_{z\to\pi} \frac{\sin z - \sin \pi}{z - \pi} = -1 \neq 0$. On the other hand, the second factor is unbounded as $z \to \pi$, namely, $\lim_{z\to\pi} \frac{1}{|z-\pi|} = \infty$. We conclude that

$$\lim_{z\to\pi} \left| \frac{\sin z}{(z-\pi)^2} \right| = \infty.$$

Thus, $z_0 = \pi$ is a pole, and not a removable singularity.

(d) The limit

$$\lim_{z\to 0} \frac{\sin z}{z^2 + z} = \lim_{z\to 0} \frac{\sin z}{z} \cdot \lim_{z\to 0} \frac{1}{z+1} = 1 \cdot 1 = 1$$

exists, $z_0 = 0$ is a removable singularity.

**Problem 38** Which of the following functions have a pole at the given point $z_0$? If $z_0$ is a pole, determine its order.

(a) $\dfrac{e^{(z^2)} - 1}{z^3}$, $z_0 = 0$.

(b) $\dfrac{z}{1 - \sin z}$, $z_0 = 0$.

(c) $\dfrac{\cos z - 1}{z^2}$, $z_0 = 0$.

(d) $\tan^2 z$, $z_0 = \frac{\pi}{2}$.

**Solution.**

(a) The power series expansion of the function $e^{(z^2)}$ is

$$e^{(z^2)} = \sum_{n=0}^{\infty} \frac{z^{2n}}{n!} = 1 + z^2 + \frac{1}{2} z^4 + \cdots ,$$

to see this, substitute $w = z^2$ in the power series of $e^w$. It follows that the limit

$$\lim_{z\to 0} \frac{e^{(z^2)} - 1}{z^2} = \lim_{z\to 0} \frac{z^2 + \frac{1}{2} z^4 + \cdots}{z^2} = 1$$

exists, and the function

$$\varphi(z) = \begin{cases} \frac{e^{(z^2)} - 1}{z^2}, & z \neq 0, \\ 1, & z = 0, \end{cases}$$

is holomorphic in $\mathbb{C}$.

For the given function we have

$$\frac{e^{(z^2)} - 1}{z^3} = \varphi(z) \frac{1}{z}, \quad \varphi(0) \neq 0.$$

It follows that $z_0 = 0$ is a simple pole of $\frac{e^{(z^2)} - 1}{z^3}$.

(b) Notice that $1-\sin z$ takes the value $1 \neq 0$ at $z_0 = 0$. Thus, $z_0 = 0$ is not a singularity of $\frac{z}{1-\sin z}$.

(c) Using the power series

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = 1 - \frac{1}{2} z^2 + \frac{1}{24} z^4 + \cdots,$$

we see that

$$\frac{\cos z - 1}{z^2} = \frac{\left(1 - \frac{1}{2} z^2 + \frac{1}{24} z^4 + \cdots\right) - 1}{z^2} = -\frac{1}{2} + \frac{1}{24} z^2 + \cdots.$$

It follows that $\lim_{z \to 0} \frac{\cos z - 1}{z^2} = -\frac{1}{2}$ exists, and therefore $z_0 = 0$ is a removable singularity and not a pole.

(d) We first consider the function $\tan z = \frac{\sin z}{\cos z}$. We have

$$\tan z = \frac{1}{z - \frac{\pi}{2}} \cdot \sin z \cdot \frac{z - \frac{\pi}{2}}{\cos z}.$$

We have $\lim_{z \to \frac{\pi}{2}} \sin z = 1$,

$$\lim_{z \to \frac{\pi}{2}} \frac{z - \frac{\pi}{2}}{\cos z} = \lim_{z \to \frac{\pi}{2}} \frac{z - \frac{\pi}{2}}{\cos z - \cos \frac{\pi}{2}} = \frac{1}{(\cos z)'|_{z = \frac{\pi}{2}}} = \frac{1}{-\sin \frac{\pi}{2}} = -1.$$

It follows that

$$\tan z = \varphi(z) \frac{1}{z - \frac{\pi}{2}}, \quad \varphi\left(\frac{\pi}{2}\right) \neq 0.$$

Hence, $z_0 = \frac{\pi}{2}$ is a simple pole of the function $\tan z$ and a double pole of the function $\tan^2 z$.

**Problem 39** Recall that we say that a function $F$ has a zero of order $n \in \mathbb{N}$ at a point $z_0$ if $F(z_0) = \cdots = F^{(n-1)}(z_0) = 0$, $F^{(n)}(z_0) \neq 0$.

Assume that $h(z) = \frac{f(z)}{g(z)}$, where the functions $f$ and $g$ are holomorphic in $D(z_0; R)$ with some $R > 0$, $f$ has a zero of order $n \in \mathbb{N}$ at $z_0$, and $g$ has a zero of order $m \in \mathbb{N}$ at $z_0$. Prove the following statements.

(a) If $n \geq m$, then $z_0$ is a removable singularity of $h$. Moreover,

$$\lim_{z \to z_0} h(z) = \frac{f^{(m)}(z_0)}{g^{(m)}(z_0)}.$$

(b) If $n < m$, then $z_0$ is a pole of $h$ of order $m - n$.

**Solution.**

Under the assumptions we have

$$f(z) = \sum_{k=n}^{\infty} a_k (z - z_0)^k = (z - z_0)^n \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k, \quad a_n \neq 0,$$
$$g(z) = \sum_{k=m}^{\infty} b_k (z - z_0)^k = (z - z_0)^m \sum_{k=0}^{\infty} b_{m+k} (z - z_0)^k, \quad b_m \neq 0.$$

(a) Suppose that $n \geq m$. Then

$$h(z) = \frac{f(z)}{g(z)} = (z - z_0)^{n-m} \frac{\sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k}{\sum_{k=0}^{\infty} b_{m+k} (z - z_0)^k},$$

where

$$\lim_{z \to z_0} \frac{\sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k}{\sum_{k=0}^{\infty} b_{m+k} (z - z_0)^k} = \frac{a_n}{b_m} \neq 0.$$

It follows that $\lim_{z \to z_0} h(z)$ exists, namely,

$$\lim_{z \to z_0} h(z) = \begin{cases} 0, & n > m, \\ \frac{a_m}{b_m}, & n = m. \end{cases}$$

Hence, $z_0$ is a removable singularity of $h$.

If $n > m$, then $f^{(m)}(z_0) = 0$. If $n = m$, then

$$\frac{a_m}{b_m} = \frac{\frac{1}{m!} f^{(m)}(z_0)}{\frac{1}{m!} g^{(m)}(z_0)} = \frac{f^{(m)}(z_0)}{g^{(m)}(z_0)},$$

and we conclude that

$$\lim_{z \to z_0} h(z) = \frac{f^{(m)}(z_0)}{g^{(m)}(z_0)}.$$

(b) Now we consider the case when $n < m$. We have

$$h(z) = \frac{f(z)}{g(z)} = \frac{1}{(z - z_0)^{m-n}} \frac{\sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k}{\sum_{k=0}^{\infty} b_{m+k} (z - z_0)^k},$$

where, again,

$$\lim_{z \to z_0} \frac{\sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k}{\sum_{k=0}^{\infty} b_{m+k} (z - z_0)^k} = \frac{a_n}{b_m} \neq 0.$$

It follows that $z_0$ is a pole of order $m - n$.

# Residues

**Problem 40** Calculate residues of given functions at all their singularities.

(a) $\dfrac{2z^2+1}{z^2+25}$,

(b) $\dfrac{1}{\sin(\pi z)}$,

(c) $\dfrac{1-\cos z}{z^3}$,

(d) $\dfrac{z^2}{(1+z)^3}$.

**Solution.**

(a) We have
$$\frac{2z^2+1}{z^2+25} = \frac{2z^2+1}{(z-5i)(z+5i)}.$$
The singularities are $5i$ and $-5i$. The residues are
$$\operatorname{Res}\left(\frac{2z^2+1}{z^2+25}, 5i\right) = \lim_{z\to 5i}(z-5i)\frac{2z^2+1}{z^2+25} = \lim_{z\to 5i}\frac{2z^2+1}{z+5i} = \frac{-49}{10i} = \frac{49}{10}i$$
and
$$\operatorname{Res}\left(\frac{2z^2+1}{z^2+25}, -5i\right) = \lim_{z\to -5i}(z+5i)\frac{2z^2+1}{z^2+25} = \lim_{z\to -5i}\frac{2z^2+1}{z-5i} = \frac{-49}{-10i} = -\frac{49}{10}i.$$

(b) Notice that $\sin(\pi z)=0$ if and only if $z=k$, $k\in\mathbb{Z}$. The singularities are $z_k=k$, $k\in\mathbb{Z}$. The residues are
$$\operatorname{Res}\left(\frac{1}{\sin(\pi z)}, k\right) = \frac{1}{(\sin(\pi z))'|_{z=k}} = \frac{1}{\pi\cos(\pi k)} = \frac{(-1)^k}{\pi}.$$

(c) The only singularity is $z_0=0$. We already have seen in Problem 38(c) that $\lim_{z\to 0}\frac{1-\cos z}{z^2} = \frac{1}{2}$. Thus,
$$\frac{1-\cos z}{z^3} = \varphi(z)\frac{1}{z}, \quad \varphi(0)\neq 0.$$
Hence, $z_0=0$ is a simple pole. Thus,
$$\operatorname{Res}\left(\frac{1-\cos z}{z^3}, 0\right) = \lim_{z\to 0} z\frac{1-\cos z}{z^3} = \lim_{z\to 0}\frac{1-\cos z}{z^2} = \frac{1}{2}.$$

(d) The only singularity is $z_0=-1$ which is a pole of order 3. We have
$$\operatorname{Res}\left(\frac{z^2}{(1+z)^3}, -1\right) = \frac{1}{2}\left((z+1)^3\frac{z^2}{(1+z)^3}\right)''\Big|_{z=-1} = \frac{1}{2}\left(z^2\right)''\Big|_{z=-1} = \frac{1}{2}\cdot 2 = 1.$$